Literacy Association of Ireland Annual Conference 2026 · AI and Literacy · Saturday 7 November

Leading Teachers Through AI

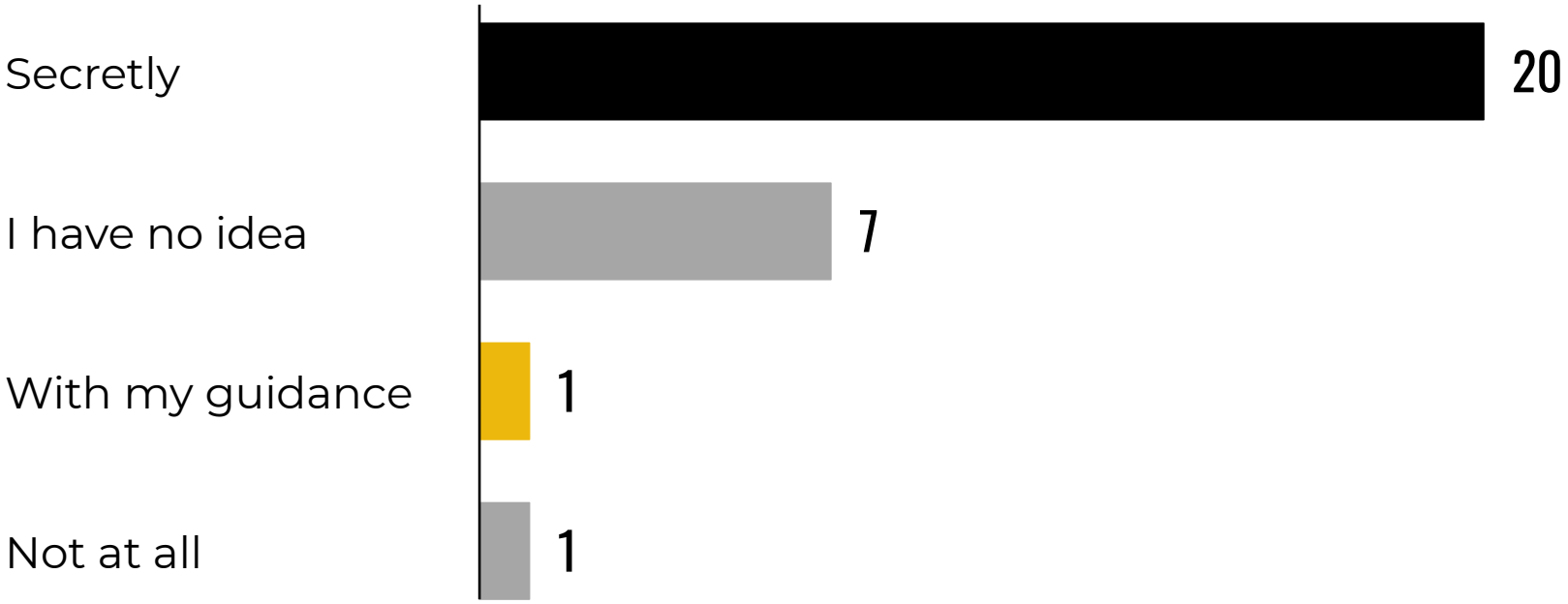
A CRAFT-framed professional development study for multiliteracy leadership

Andrea C. Borowczak, Ed.D.
Professor and Director, School of Teacher Education

Mike Borowczak, Ph.D.
Associate Professor of Electrical and Computer Engineering

University of Central Florida

“How are your students using AI right now?”



1 of 29

said students use AI with the teacher's guidance

> “I don't know how to spot AI in their work.”
>
> High school mathematics teacher

Opening activity, Workshop 2, 18 April 2026 · 29 named responses from K–12 teachers in the Western Regional Noyce Network

What worried them

> “AI thinks for them.”

High school biology teacher

> “That they take the responses as gospel.”

Kindergarten teacher

> “Students no longer having a stake in their learning.”

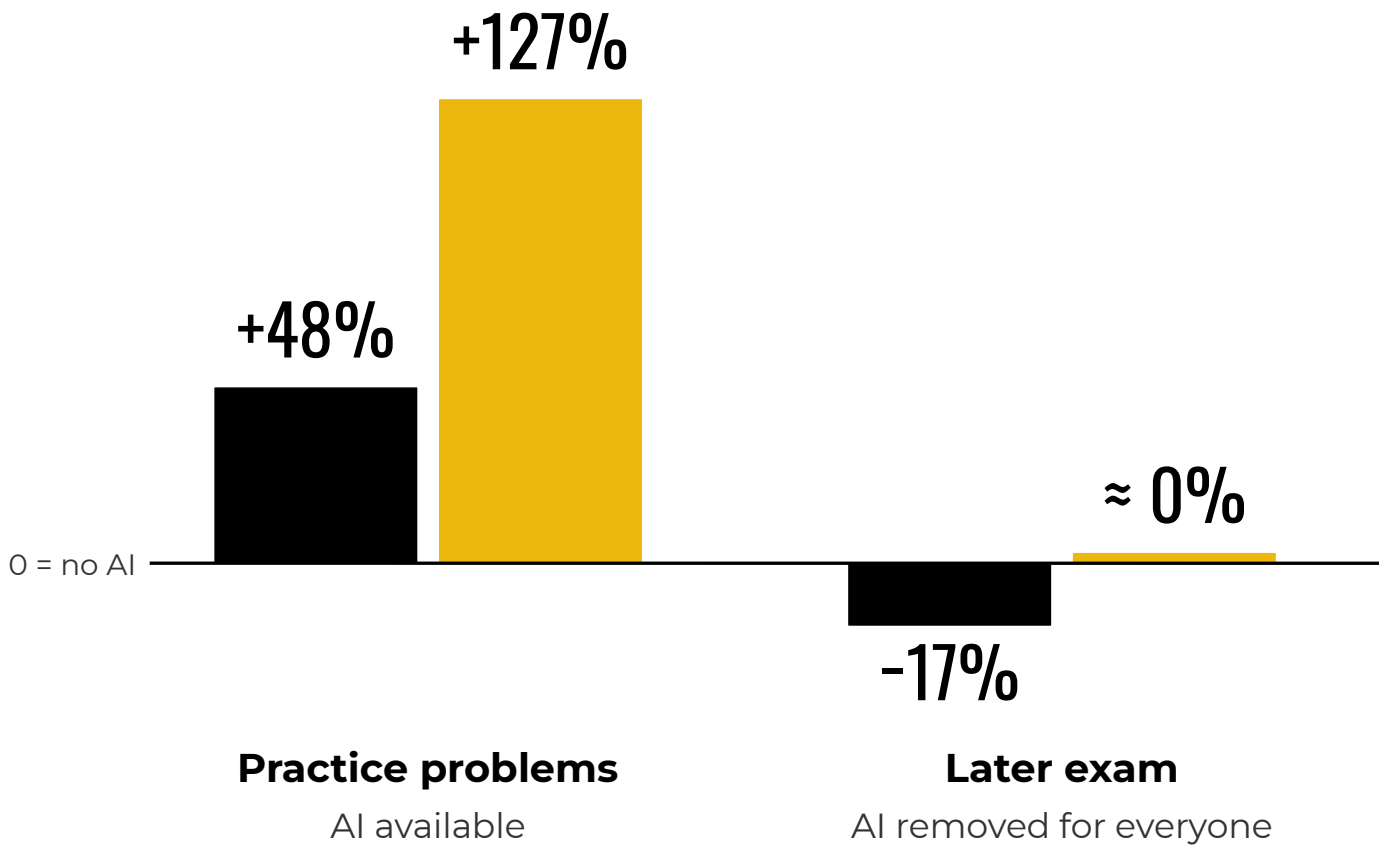
Middle school science teacher

- Most worries named lost thinking: copying, shortcuts, no struggle
- Few named whether the AI's answer was right

“Your biggest worry about student AI use,” Workshop 2 opening activity, 18 April 2026 · 26 named responses

Practice scores rose with AI; exam scores without it fell

Change in grade compared with students who had no AI



Two chatbots, both GPT-4

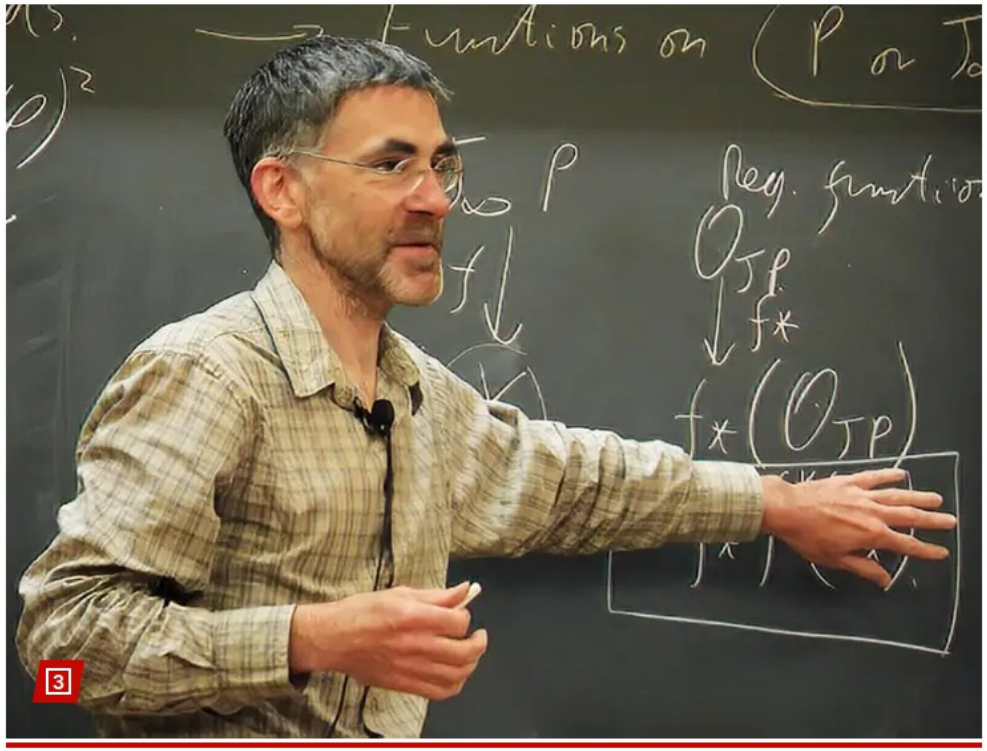
Standard chatbot
gives answers on request

Tutor chatbot
gives hints first, holds back answers

With answers held back, exam scores held steady.

Bastani et al. (2025), PNAS 122(26) · randomized field experiment, nearly 1,000 high school mathematics students in Turkey

A Fields Medalist's answer: exams with no devices



Richard Borcherds, a mathematician at UC Berkeley, is pictured.
Instagram/@quantumaths

IN THE NEWS **The California Post · 25 October 2026**

> "Online exams are now a waste of time as several students just use AI to answer the questions. I cannot tell which students are using AI, but I notice the average score increases a lot in online exams."

Richard Borcherds, UC Berkeley, Fields Medal 1998

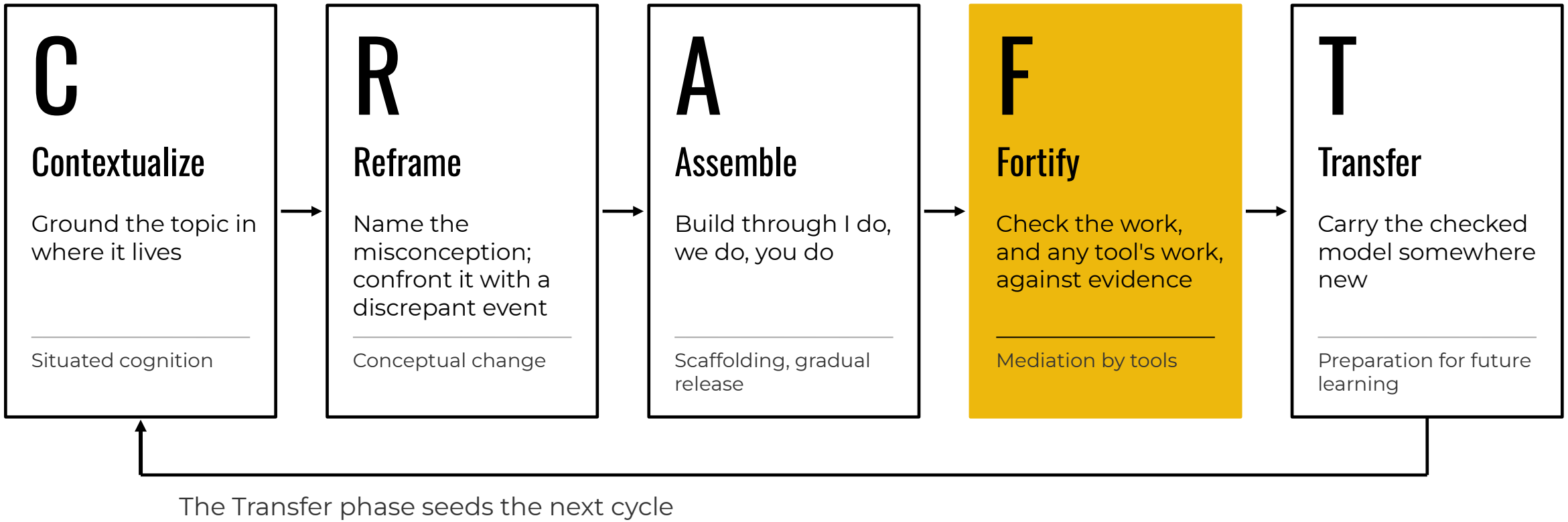
- Stopped grading homework for correctness
- Now uses in-person exams with no devices

The California Post, 25 October 2026 · photo: Instagram/@quantumaths, via The California Post

CRAFT · Contextualize · Reframe · Assemble · Fortify · Transfer

Build understanding. Trust nothing without evidence.

Five phases, each defined by what it does



The Fortify phase runs in every cycle, with its own routine: Check the Machine

Two workshops, one week apart

Workshop 1 · 11 April 2026

Using AI for STEM

- All five phases, modeled through lesson planning
- One prompt, three chatbots, compared in groups of three or more
- Each teacher builds an AI-assisted lesson for their own standard

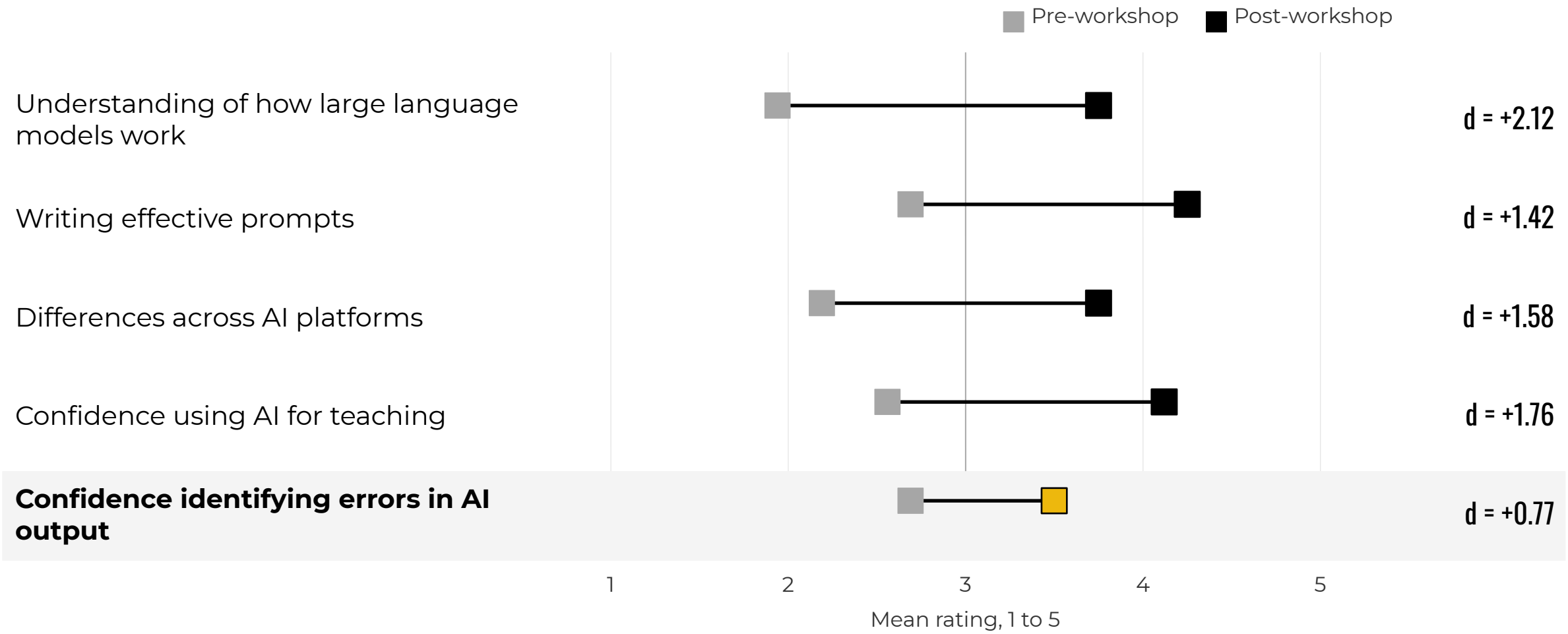
Workshop 2 · 18 April 2026

Checking STEM AI Outputs

- Built around the Fortify phase and its Check the Machine routine
- Error hunts in code and content; break an AI in your own subject
- Classroom norms, then a classroom version of the routine

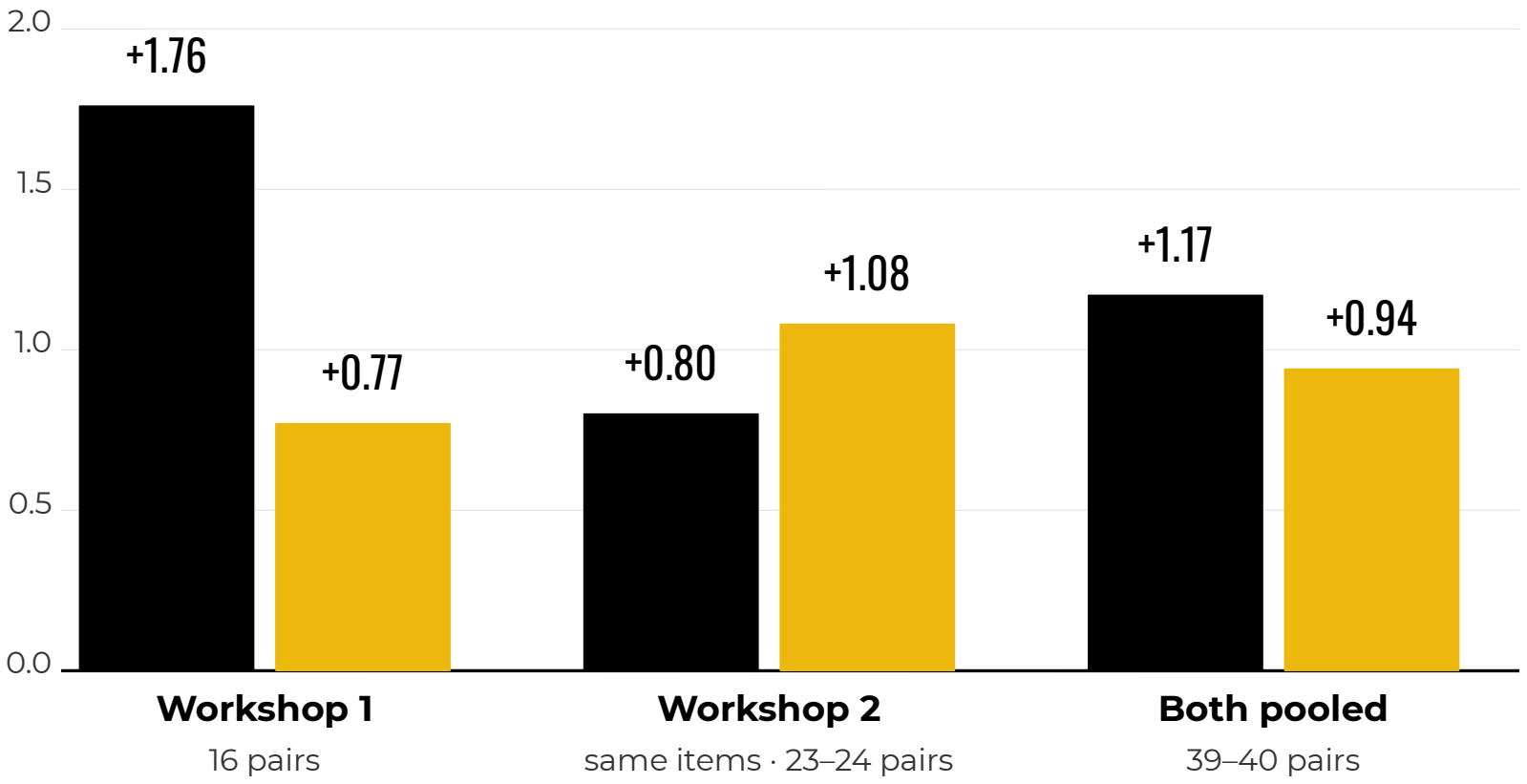
- K–12 educators in the Western Regional Noyce Network: in-service and pre-service teachers, Noyce Scholars and Fellows
- Online, half a day each · surveys before and after each day · 16 and 24 matched pairs

One workshop raised every item; checking rose least



Workshop 1, 16 matched pairs · Wilcoxon signed-rank p < .02 for every item · d: Cohen's d for repeated measures · preliminary, manuscript under review

A second session built on checking moved checking further



Confidence using AI

Confidence identifying errors

Effect size d, same wording in both workshops

The order flipped after the checking session

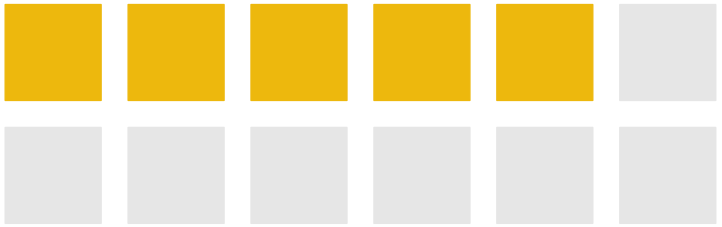
Five teachers came to both: they held their gains, then rose further

Workshop 2 repeated the Workshop 1 confidence items word for word · two sessions, not a controlled dose-response design · preliminary

Their own writing shows the same gap

Workshop 1 lesson ideas: “How students verify”

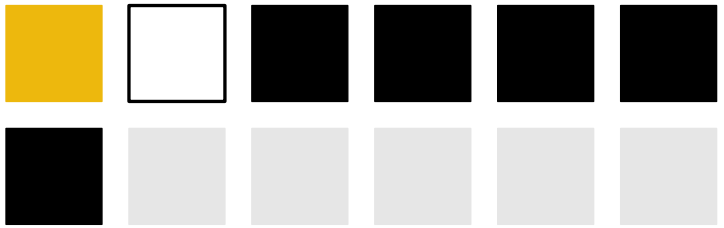
5 of 12 named a specific check on the AI's output



- Compare with the textbook; graph it by hand; lateral reading
- A use, a check on the learner, or “dig deeper”

Workshop 2 norms, asked to require checking

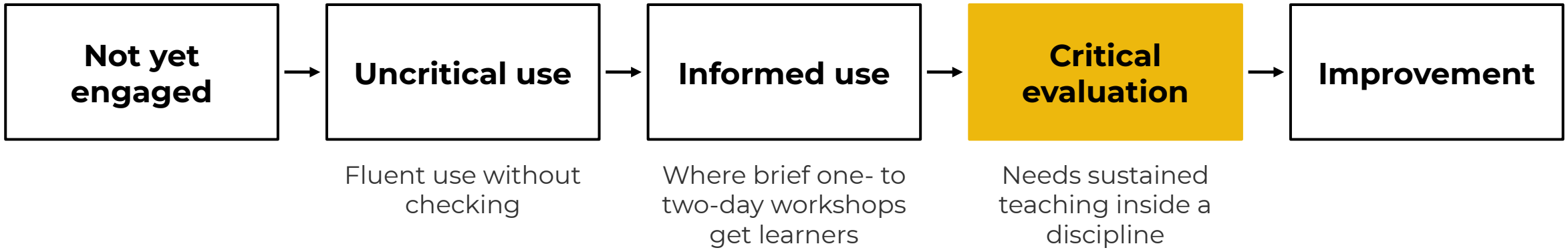
1 of 12 asked students to check what the AI said



- “Filter information: use a rubric and cite source”
- Write your answer before you type it
- Disclose or cite AI use (5)
- Limits, privacy, and uses (5)

Shared workspaces, Workshop 1 (11 April) and Workshop 2 (18 April, before the Fortify-phase activities) · first-pass coding by the presenters

Others report the same shape



Five-stage continuum of AI literacy, Liu and Levy (2026) · North Carolina State University, more than 330 participants, Fall 2024 to Spring 2026

- Their evidence is instructor observation and artifact review; ours is matched surveys: different methods, the same lag
- The American Association of Colleges for Teacher Education (2026) asks candidates to audit every AI-generated lesson plan and to train against automation bias

For those who lead teacher learning

1

Expect the gap after one workshop

Vocabulary and confidence arrive in a day; plan for the checking habit to lag

2

Follow with short, repeated checking cycles

In groups of three or more first, then alone, on material from each teacher's own subject

3

Name the checking phase

A named phase gets time on the schedule and something to measure

What these data cannot tell us yet

- Small samples from one network: 16 and 24 matched pairs, five teachers in both
- Self-reported confidence; we did not measure how many errors teachers caught
- A survey that has not been independently validated
- Two workshops, not a controlled dose-response design

Next

A detection task: artifacts with known errors, scored by what teachers catch

Next: Check the Machine, the routine inside the Fortify phase

One session teaches people to use AI.

Repeated, structured checking teaches them when to trust it.

Andrea C. Borowczak · Andrea.Borowczak@ucf.edu
Mike Borowczak · mike.borowczak@ucf.edu

Workshop materials, openly licensed: cxedhub.com